Project Submission & Architecture Report

Technical Domain — Round 2 Task: Personal Portfolio with Integrated AI Chatbot

Candidate Name: Siddhartha Bolla
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Program: B.Tech in CSE (AI & ML) [2025–2029]
Current CGPA: 9.45 / 10

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📌 1. Executive Summary & Objective

This project was engineered to fulfill the requirements of the **Technical Domain — Round 2 Task**. The objective is to present a high-performance personal portfolio website coupled with an intelligent, profile-aware AI conversational chatbot capable of answering questions about Siddhartha's background, education, technical skills, projects, resume, and coding statistics.

A critical architectural priority was **universal cross-device portability**: the repository is completely self-contained with **zero external server processes, zero compilation steps, and zero mandatory API keys**. Whoever clones or opens the repository on their computer can run and evaluate it immediately.

🎯 2. Requirements Compliance Matrix

Task Specification	Implementation Details	Evaluation Status
Personal Portfolio	Modern, responsive glassmorphic UI with dark/light mode toggle representing Siddhartha Bolla, education at VNR VJIET (CGPA: 9.45, Batch 2025–2029), skills, and projects.	✅ Completed
Integrated AI Chatbot	Self-contained conversational agent with multi-intent NLP engine, quick suggestion chips, voice input (speech-to-text), voice synthesis (read aloud), and dynamic action cards.	✅ Completed
Resume Accessibility	Dual accessibility: Full-screen interactive online resume modal + direct PDF download button linking to `assets/resume.pdf`.	✅ Completed
Coding Profiles	Integrated stats and direct clickable badges for LeetCode (200+ solved), GitHub (siddharthabolla456-tech), HackerRank (5-Star), and CodeChef.	✅ Completed
Universal Portability	Runs on any PC (Windows, Mac, Linux) by double-clicking `index.html`, launching `start.bat`, or running Python/Node local servers.	✅ Completed
Public Hosting Ready	Automated GitHub Pages workflow (`.github/workflows/deploy.yml`) pre-configured for 1-click global deployment.	✅ Completed

🤖 3. AI Chatbot Technical Architecture

The chatbot is built to deliver instant, contextual responses without relying on paid cloud APIs that might fail or expire during evaluation:

- **Client-Side NLU & Token Matcher:** Employs weighted token scoring, entity extraction, and synonym expansion. Handles varied phrasings including "*What projects has he worked on?*", "*What technologies does he know?*", "*Show me his resume*", "*What are his coding profiles?*", and "*What is his CGPA?*".
- **Interactive Action Triggers:** Bot responses render interactive HTML action cards with clickable buttons that directly launch the Resume Modal, initiate PDF downloads, or open verified LeetCode and LinkedIn profiles.
- **Zero-Dependency Audio Synthesis:** Generates responsive sound effects using the browser's native Web Audio API oscillators without requiring external MP3 downloads.
- **Speech Recognition & Synthesis:** Supports hands-free voice input via the Web Speech API (`webkitSpeechRecognition`) and text-to-speech narration (`speechSynthesis`).
- **Dual Mode (Offline + Gemini Live):** Operates offline by default; includes an optional settings modal where evaluators can enter a Google Gemini API key to activate live LLM generative chat.

🚀 4. Featured Projects Breakdown

1. KrishiLink AI — Smart Agri-Commerce & Predictive ML Ecosystem

React • FastAPI • ML • Node.js

Agricultural Marketplace with Machine Learning Crop Forecasting

An end-to-end full-stack platform bridging smallholder farmers, Farmer Producer Organizations (FPOs), and institutional buyers. Features machine learning services built in FastAPI and Scikit-Learn to forecast crop yield and market commodity rates, combined with interactive Leaflet map trade matchmaking.

2. Smart Feedback Management System

Node.js • Express • MongoDB • REST APIs

College Hackathon Project — Academic Feedback Analytics

Scalable RESTful backend service engineered for college hackathon competition. Implements secure JWT authentication, role-based access control (RBAC) across student, faculty, and administrative tiers, automated sentiment aggregation, and modular Postman test suites.

3. AI Portfolio with Integrated Conversational Agent

JavaScript • NLP Engine • Web Audio • Glassmorphism

Portable Personal Web Application with Interactive AI

A high-performance personal portfolio featuring this autonomous AI chatbot. Designed with dark/light theme persistence, dynamic typing effects, animated statistics counters, full resume inspection modal, and cross-device zero-dependency portability.

💻 5. Technical Skills Summary

Domain	Technologies & Frameworks
Programming Languages	Python, JavaScript (ES6+), C++, C, SQL, HTML5, CSS3
Frontend & Web	React.js, Tailwind CSS, Vite, Vanilla JavaScript, Responsive Design
Backend & Databases	Node.js, Express.js, FastAPI, RESTful APIs, JWT Auth, MongoDB, MySQL
AI / Machine Learning	Scikit-Learn, Pandas, NumPy, Predictive Modeling, Exploratory Data Analysis
Developer Tools	Git, GitHub, Postman, Docker, VS Code, Linux / Bash

⚡ 6. How to Run & Deploy the Project

The project provides four straightforward execution methods:

Method A: Instant Browser Launch (Zero Installation)

Double-click `index.html` (or `start.bat` on Windows) in file explorer. The application opens directly in any modern web browser.

Method B: Python Local Server

```
python server.py
# Server opens automatically at http://localhost:8000
```

Method C: Node.js / NPM

```
npm start
# or: npx serve .
```

Method D: 1-Click Public Deployment (GitHub Pages)

Push the repository to GitHub and enable GitHub Pages under **Settings ➔ Pages ➔ Source: GitHub Actions**. GitHub automatically builds and hosts the site at:

`https://siddharthabolla456-tech.github.io/siddhartha-portfolio-chatbot/`

📞 7. Task Submission & Contacts

Evaluation Contacts (Round 2 Coordinators):

- **Pallavi (Technical Head):** +91 8919505048
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